

Chapter 155 — Prostate Biopsy: Techniques and Imaging — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Gold standard (chapter):** systematic **12-core** transrectal ultrasound (TRUS) biopsy $\pm$ targeting; **6 cores inadequate**.
- **European Association of Urology (EAU) 2023:** transperineal preferred over transrectal (strong).
- **Prostate Cancer Prevention Trial (PCPT):** no safe PSA threshold; PSA $\leq 4 \rightarrow$ 15% cancer.
- **PSA density (PSAD) >0.15 ng/mL/cc $\rightarrow$ suggests malignancy.**
- **39% of cancers isoechoic;** hypoechoic lesion not pathognomonic.
- **Transrectal prophylaxis for all;** ≥ 1 h before; ≤ 24 h.
- **Post-biopsy sepsis $\rightarrow$ carbapenem;** avoid empirical fluoroquinolone.
- **PROMIS:** MRI sensitivity **93% vs TRUS 48%; 27% avoid biopsy.**
- **PRECISION:** MRI-targeted **38% vs 26%** significant cancer; 28% avoid biopsy.
- **PIRADS:** **DWI dominates PZ; T2WI dominates TZ;** tie-breakers only at 3.

ULTRASONOGRAPHIC ANATOMY

- **Zones not visible** on TRUS; PZ/CZ echogenic, homogeneous vs TZ heterogeneous.
- **Corpora amylacea:** surgical-capsule calcification marking PZ-TZ plane; age-related, normal.
- **Normal seminal vesicle (SV):** $4.5\text{--}5.5 \times 2$ cm.
- **Solid SV mass $\rightarrow$ likely malignant;** cystic $\rightarrow$ benign. Endemic area $\rightarrow$ schistosomiasis.
- **Absent SV $\rightarrow$ ipsilateral renal agenesis 79%.**

GREY-SCALE TRUS

- **Probes 6–10 MHz;** end- vs side-fire: **no detection difference.**
- **7 MHz:** focal range **1–4 cm**, best for PZ. **4 MHz:** 2–8 cm, better volume.
- **Gain reference:** normal PZ = medium grey.
- **Positions:** transrectal $\rightarrow$ left lateral decubitus; transperineal $\rightarrow$ lithotomy.
- **Side-fire rotation:** clockwise $\rightarrow$ left lobe; anticlockwise $\rightarrow$ right.
- **End-fire, left lateral decubitus:** handle down $\rightarrow$ right; handle up $\rightarrow$ left.
- **Handle toward scrotum $\rightarrow$ cephalad;** toward sacrum $\rightarrow$ caudal.
- **Stepper:** rotation only.
- **Ellipsoid** $\pi/6 \times T \times AP \times L$; **sphere** $\pi/6 \times T^3$; **prolate spheroid** $\pi/6 \times T^2 \times AP$.
- **All formulas $r > 0.90$; $1\text{ cm}^3 \approx 1\text{ g}$.**
- **Measure AP sagittally;** axial AP overestimates.
- **Normal 20–25 g** until ~ 50 ; **$\approx 48\text{ g}$ at 60–70.**
- **Planimetry (3–5 mm steps) = most accurate $\rightarrow$ brachytherapy.**
- **Utricle:** midline; **hypospadias commonest** association.
- **Müllerian duct cyst:** midline, pear-shaped, neck to verumontanum $\rightarrow$ check kidneys.
- **Ejaculatory duct cyst:** off midline $\rightarrow$ azoospermia.
- **SV cyst $\rightarrow$ renal agenesis up to two-thirds.**
- **Acquired TZ cyst = haemorrhagic BPH nodule.**
- **Hyperechoic cancer $\leq 1\%$.**
- **Commonest false-positive hypoechoic:** BPH, prostatitis.
- **Extraprostatic extension (EPE):** focal loss of bright periprostatic fat.
- **Androgen deprivation:** $\approx 30\%$ volume $\downarrow$ (10–60%).
- **5 α -reductase inhibitor:** $\approx 21\%$ $\downarrow$ at 6 months.
- **Brachytherapy:** $\sim 1/2$ $\downarrow$ at 1 y, $\sim 1/2$ by 8 y.
- **Seeds:** hyperechoic with shadowing; periurethral sparing.
- **Post-prostatectomy recurrence:** hypoechoic in 75–95%.
- **Post-prostatectomy normal:** smooth bladder-neck taper; anterior nodule = dorsal vein ligature.

- **Blind anastomotic biopsy (PSA only) $\rightarrow$ uninformative.**
- **Urothelial carcinoma:** stromal lesions hypoechoic in 71%.
- **Hypoechoic lesion after BCG:** granuloma vs tumour $\rightarrow$ biopsy.
- **Squamous cell carcinoma (SCC):** anterior hyperechoic mass.
- **Adenoid cystic:** multiple small anechoic cysts.
- **Rhabdomyosarcoma:** isoechoic.
- **Lymphoma:** invisible.

PROSTATE BIOPSY: TECHNIQUES AND OUTCOMES

- **Initial biopsy yield $\approx 30\%$. Core biopsy = technique of choice;** fine-needle aspiration obsolete.
- **ERSPC benefit vs PLCO none;** lead-time reanalysis: both compatible with benefit.
- **AUA/SUO 2023:** baseline PSA 45–50; every 2–4 y at 50–69.
- **AUA/SUO 2023, high risk:** germline mutation, family history, African ancestry $\rightarrow$ start 40–45.
- **USPSTF 2018:** 55–69 individual decision.
- **Elevated-risk PSA:** >1 ng/mL at 40; >2 ng/mL at 60.
- **NCCN biopsy trigger:** abnormal DRE + raised PSA.
- **NCCN biopsy trigger:** repeat PSA >3 + PIRADS ≥ 3 .
- **Not alone:** PSA velocity; abnormal DRE with normal PSA.
- **Suspicious nodule:** 5–30% cancer risk.
- **ASAP / atypical intraductal proliferation:** 50% $\rightarrow$ repeat biopsy.
- **Absolute contraindications:** significant coagulopathy; severe immunosuppression; acute UTI or prostatitis.
- **No anus $\rightarrow$ in-bore MRI or transperineal biopsy.**
- **Anal stenosis or pain $\rightarrow$ general or regional anaesthesia.**
- **Low-dose aspirin:** continue.
- **Warfarin, clopidogrel:** stop 5–7 days.
- **Direct oral anticoagulants (DOACs):** stop 2–5 days.
- **International normalised ratio (INR) <1.5 before biopsy.**
- **Bridging:** mechanical valve; rivaroxaban.
- **Class III (contaminated) procedure.**
- **AUA 2019 prophylaxis:** fluoroquinolone; 1st/2nd-gen cephalosporin $\pm$ aminoglycoside; 3rd-gen cephalosporin.
- **Route:** oral only for quinolones; intramuscular (IM) for all.
- **Single-dose agents:** levofloxacin 500 mg; gentamicin 5 mg/kg.
- **Single-dose agents:** amikacin 15 mg/kg; ceftriaxone 1–2 g.
- **Renal impairment:** aztreonam replaces aminoglycoside.
- **Endocarditis or prosthesis risk:** IV ampicillin (vancomycin) + gentamicin $\rightarrow$ 2–3 days fluoroquinolone.
- **Infection risk factors:** antibiotics <6 months = most consistent $\rightarrow$ augmented or targeted prophylaxis.
- **Charlson score >1 = infection risk factor.**
- **Periprostatic block:** 1–2% lidocaine, 7-inch 22G.
- **Periprostatic block:** 5 mL each side at SV–prostate junction.

- **Saturation biopsy:** ≤22 mL of 1% lidocaine.
- **Topical rectal anaesthetic:** inferior to periprostatic block.
- **Needle:** 18G gun; throw 15–25 mm; core 1.0–1.5 cm.
- **Tip overshoot:** 0.5 cm beyond core → indent capsule, withdraw, fire.
- **Specimen jars:** ≤2 cores per jar.
- **Hodge 1989:** sextant; parasagittal cores miss posterolateral PZ, where most cancers arise.
- **6 → 12 cores:** ↑ detection.
- **12 → 18–21 cores (initial):** no ↑ detection.
- **TZ cores** if gland >50 mL or repeat.
- **SV biopsy:** palpable abnormality or PSA >30.
- **Solitary TZ cancer** <5% → transurethral biopsy obsolete.
- **ERSPC repeat biopsies:** 22 → 10 → 5 → 4%.
- **3rd/4th biopsy:** only if high suspicion; target the apex.
- **Serious complication** <2%.
- **30-day admission:** 1.0 → 4.1% (mostly infection).
- **Haematuria** 23–63%; clot retention 0.7%.
- **Rectal bleeding** 2.1–21.7%.
- **Haematospermia** 9.8–50.4% (~1 month).
- **Infection** 0.1–7%; **sepsis** 0.3–3.1%.
- **Infective admission** 0.6–4.1%; **UTI** 2–6%.
- **Vasovagal** 1.4–5.3% → Trendelenburg + IV fluids.
- **Retention** 0.2–0.4% (large gland, high IPSS).
- **Sepsis:** carbapenem; up to 2 weeks antibiotics if admitted.
- **E. coli resistant:** fluoroquinolone, trimethoprim-sulfamethoxazole (TMP-SMX), ampicillin.
- **E. coli susceptible:** 2nd/3rd-gen cephalosporins, amikacin, carbapenems.
- **Rectal bleed escalation:** pressure → balloon tamponade → endoscopic adrenaline/sclerosant → embolisation → suture.

ADVANCED AND INVESTIGATIONAL TECHNIQUES

- **Colour Doppler:** frequency shift, direction-dependent; red toward, blue away.
- **Power Doppler:** amplitude; slow flow; direction-independent.
- **Doppler vs grey scale:** neither Doppler mode superior; neither replaces systematic biopsy.
- **Tumour microvessels** 10–15 μm: invisible without contrast.
- **Microbubbles:** 1–10 μm.
- **Flash replenishment:** high-power flash destroys bubbles → low-power refill imaged.
- **Elastography:** stiffness; complements, never replaces, systematic biopsy.
- **Micro-ultrasound:** 29 MHz, 70 μm; PRI-MUS score.
- **Micro-ultrasound:** no contrast → renal failure.
- **PIRADS v1:** ESUR 2012. **PIRADS v2:** ACR + ESUR + AdMeTech 2015.
- **PIRADS significant cancer:** Gleason ≥7 / >0.5 cm³ / EPE.
- **PZ:** DWI 3 + DCE positive → 4.
- **TZ:** T2WI 3 + DWI 5 → 4.
- **Automatic PIRADS 5:** >1.5 cm or EPE.
- **PIRADS cancer rates:** 3 ≈ ⅓, 4 >½, 5 ≈ 90%.

- **PROMIS:** 576 men; template mapping reference.
- **PROMIS definition:** significant = Grade Group ≥3 or core ≥6 mm.
- **PROMIS:** NPV 89 vs 74%; +18% significant cancer with targeting.
- **PRECISION:** 500 men, randomised non-inferiority; targets PIRADS ≥3.
- **MRI false negatives:** ≈⅓ of misses primary pattern 4.
- **In-bore vs cognitive vs fusion:** no accuracy difference (meta-analysis).
- **Cognitive targeting:** 2–4 cores per target + systematic.
- **AUA/SUO 2023:** target any PIRADS ≥3 lesion; 2–3 cores per target.
- **AUA/SAR 2016:** MRI after negative biopsy + rising PSA.
- **UroNav:** freehand, electromagnetic tracking.
- **Artemis:** mechanical arm.
- **Transperineal perineum:** scrotum taped up; perineum on tension.
- **Periapical triangle:** levator ani (lateral), rhabdosphincter (medial), external anal sphincter (posterior), apex (superior).
- **Periapical block:** 1–2 mL skin → 5 mL triangle → 3 mL capsule.
- **Transperineal pain:** VAS <4; injection hurts most.
- **Freehand learning curve:** 4–6 weeks; 12–14 min.
- **Transperineal access:** two 14G cannulas cheapest; PrecisionPoint commonest kit.
- **Transperineal infection:** lower than transrectal.
- **Transperineal antibiotics:** skin-flora only or none.
- **Transperineal bleeding:** no rectal bleeding.
- **Modified Barzell:** 20 cores.
- **Ginsburg:** adds base sector if >50 g.
- **MUSIC template:** 12-core parity; no midline cores.
- **Anterior-only cancers:** 6–20% → transperineal reaches them.
- **Barringer 1922:** first transperineal biopsy.

CLASSIC TRAPS

- **Trap:** no hypoechoic lesion → no biopsy — **Actually:** 39% isoechoic; biopsy anyway.
- **Trap:** PSA ≤4 is safe — **Actually:** PCPT, no safe threshold.
- **Trap:** infection commonest complication — **Actually:** bleeding commonest; infection commonest serious.
- **Trap:** saturation beats 12-core initially — **Actually:** no gain; reserve for repeat.
- **Trap:** oral cephalosporin for transrectal prophylaxis — **Actually:** oral only for quinolones.
- **Trap:** stop aspirin — **Actually:** low-dose aspirin continues.
- **Trap:** DCE matters in TZ — **Actually:** TZ uses DWI tie-breaker; DCE only PZ 3 → 4.
- **Trap:** renal agenesis mainly with müllerian cysts — **Actually:** wolffian SV cysts.
- **Trap:** fire after passing capsule — **Actually:** 0.5 cm overshoot samples TZ; withdraw first.
- **Trap:** empirical ciprofloxacin for post-biopsy sepsis — **Actually:** carbapenem.